

Let $X \sim p_0(\mathbf{x})$ be the target distribution, $Y_\sigma = \mathbf{x}_0 + \sigma \mathbf{Z}$

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The conditional DSM loss is:

$$\mathcal{L}_{\sigma,U} = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), U} \|\mathbf{D}_\theta(\mathbf{x}_0 + \sigma \mathbf{Z}; \sigma, U) - \mathbf{x}_0\|_2^2$$

The unconditional DSM loss is:

$$\mathcal{L}_\sigma = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{D}_\theta(\mathbf{x}_0 + \sigma \mathbf{Z}; \sigma) - \mathbf{x}_0\|_2^2$$

where U satisfies the Markov condition $U \rightarrow X \rightarrow Y_\sigma$, i.e. $U \perp Y_\sigma \mid X$. Note that from Proposition 7, for any denoiser $\mathbf{D}_\theta$,

$$\mathcal{L}_\sigma \geq \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{D}^*(\mathbf{y}; \sigma) - \mathbf{x}_0\|_2^2 = \text{MMSE}[X \mid Y_\sigma]$$

and

$$\mathcal{L}_{\sigma,U} \geq \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), U} \|\mathbf{D}^*(\mathbf{y}; \sigma, U) - \mathbf{x}_0\|_2^2 = \text{MMSE}[X \mid Y_\sigma, U]$$

Recall that:

$$I(X, U) = \frac{1}{2} \int_0^\infty \frac{dt}{t^2} (\text{MMSE}[X \mid Y_t] - \text{MMSE}[X \mid Y_t, U])$$

For now treat $\mathcal{L}_{\sigma,U}$ and $\mathcal{L}_\sigma$ as proxies for $\text{MMSE}[X \mid Y_t]$ and $\text{MMSE}[X \mid Y_t, U]$ respectively (since we can't assign an inequality sign to $\text{MMSE}[X \mid Y_t] - \text{MMSE}[X \mid Y_t, U] \stackrel{?}{>} \mathcal{L}_{\sigma,U} - \mathcal{L}_\sigma$; though for the Gaussian case these 2 quantities are the same). Consider the case of a linear denoiser: $\mathbf{D}(\mathbf{y}; \sigma) = \mathbf{W}_\sigma \mathbf{y} + \mathbf{b}_\sigma$. Then:

$$\mathcal{L}_\sigma = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \quad (1.1)$$

We want a closed-form expression for $\mathcal{L}_\sigma$ (if exists). Recall that the denoiser is learned such that the DSM loss is minimized, so we can solve for the optimal weights $\mathbf{W}_\sigma$ and $\mathbf{b}_\sigma$. First take the derivative of $\mathcal{L}_\sigma$ w.r.t. $\mathbf{b}_\sigma$ and set to 0; note that

$$\nabla_{\mathbf{b}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 = 2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)$$

So $\|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$ is continuous, and its partial derivative w.r.t. $\mathbf{b}_\sigma$ exists, and $\mathbb{E}[2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)]$ is bounded if $\mathbb{E}[\mathbf{x}_0]$ is bounded, i.e. the first moment of p_0 is finite; in which case we can apply Leibniz's rule and push the differentiation inside the expectation:

$$\begin{aligned} & \nabla_{\mathbf{b}_\sigma} \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\nabla_{\mathbf{b}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2] \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)] = 2(\mathbb{E}_{\mathbf{x}_0 \sim p_0}[(\mathbf{W}_\sigma - \mathbf{I})\mathbf{x}_0] + \mathbf{b}_\sigma) \end{aligned}$$

Setting the above to 0 and solve for $\mathbf{b}_\sigma$:

$$\mathbf{b}_\sigma = (\mathbf{I} - \mathbf{W}_\sigma) \mathbb{E}[\mathbf{x}_0]$$

Now with optimal $\mathbf{b}_\sigma$ found, take the derivative of $\mathcal{L}_\sigma$ w.r.t. $\mathbf{W}_\sigma$ and set to 0; similarly we note:

$$\nabla_{\mathbf{W}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 = 2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)(\mathbf{x}_0 + \sigma \mathbf{Z})^\top$$

So $\|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$ is continuous and its partial derivative w.r.t. $\mathbf{W}_\sigma$ exists, and $\mathbb{E}[2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)(\mathbf{x}_0 + \sigma \mathbf{Z})^\top]$ is bounded if $\mathbb{E}[\mathbf{x}_0 \mathbf{x}_0^\top]$ is bounded, i.e. the second moment of p_0 is finite; in which case we can apply Leibniz's rule and push the differentiation inside the expectation:

$$\begin{aligned} & \nabla_{\mathbf{W}_\sigma} \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\nabla_{\mathbf{W}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2] \end{aligned} \quad (1.2)$$

$$\begin{aligned} &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0)(\mathbf{x}_0 + \sigma \mathbf{Z})^\top] \\ &= 2(\mathbf{W}_\sigma \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})}[(\mathbf{x}_0 + \sigma \mathbf{Z})(\mathbf{x}_0 + \sigma \mathbf{Z})^\top] + \mathbf{b}_\sigma \mathbb{E}[\mathbf{x}_0^\top] - \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})}[\mathbf{x}_0(\mathbf{x}_0 + \sigma \mathbf{Z})^\top]) \\ &= 2(\mathbf{W}_\sigma (\mathbb{E}_{\mathbf{x}_0 \sim p_0}[\mathbf{x}_0 \mathbf{x}_0^\top] + \sigma^2 \mathbf{I}) - \mathbb{E}_{\mathbf{x}_0 \sim p_0}[\mathbf{x}_0 \mathbf{x}_0^\top] + \mathbf{b}_\sigma \mathbb{E}[\mathbf{x}_0]^\top) \end{aligned} \quad (1.3)$$

$$\begin{aligned} & \text{(since } \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})}[\mathbf{Z} \mathbf{Z}^\top] = \mathbf{I} \text{ and as } \mathbf{Z} \perp \mathbf{x}_0 \\ & \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})}[\mathbf{x}_0 \mathbf{Z}^\top] = \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})}[\mathbf{Z}^\top] \mathbb{E}[\mathbf{x}_0] = \mathbf{0} \mathbb{E}[\mathbf{x}_0] = \mathbf{0}) \end{aligned}$$

Substituting $\mathbf{b}_\sigma = (\mathbf{I} - \mathbf{W}_\sigma)\mathbb{E}[\mathbf{x}_0]$ and setting (1.3) to 0:

$$\begin{aligned} \mathbf{W}_\sigma (\mathbb{E}[\mathbf{x}_0\mathbf{x}_0^\top] + \sigma^2\mathbf{I}) - \mathbb{E}[\mathbf{x}_0\mathbf{x}_0^\top] + (\mathbf{I} - \mathbf{W}_\sigma)\mathbb{E}[\mathbf{x}_0]\mathbb{E}[\mathbf{x}_0]^\top &= \mathbf{0} \\ \mathbf{W}_\sigma (\mathbb{E}[\mathbf{x}_0\mathbf{x}_0^\top] + \sigma^2\mathbf{I} - \mathbb{E}[\mathbf{x}_0]\mathbb{E}[\mathbf{x}_0]^\top) - \mathbb{E}[\mathbf{x}_0\mathbf{x}_0^\top] + \mathbb{E}[\mathbf{x}_0]\mathbb{E}[\mathbf{x}_0]^\top &= \mathbf{0} \end{aligned}$$

Note that $\mathbb{E}[\mathbf{x}_0\mathbf{x}_0^\top] = \Sigma_{p_0} + \mu_{p_0}\mu_{p_0}^\top$ where $\mu_{p_0} = \mathbb{E}[\mathbf{x}_0]$. then solve for $\mathbf{W}_\sigma$:

$$\mathbf{W}_\sigma = \Sigma_{p_0}(\Sigma_{p_0} + \sigma^2\mathbf{I})^{-1} \quad (1.4)$$

Now substitute (1.4) back into (1.1), we get:

$$\begin{aligned} \mathcal{L}_\sigma &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z}) + (\mathbf{I} - \mathbf{W}_\sigma)\mu_{p_0} + \sigma\mathbf{Z}\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}) + \sigma\mathbf{Z}\|_2^2 \end{aligned} \quad (1.5)$$

$$= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [((\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}))^\top (\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})] \quad (1.6)$$

$$+ 2((\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}))^\top \sigma\mathbf{Z} + (\sigma\mathbf{Z})^\top (\sigma\mathbf{Z})] \quad (1.7)$$

We consider each of the 3 expectation terms, firstly:

$$\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [((\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}))^\top (\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})] \quad (1.8)$$

$$= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\text{Tr}((\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}))^\top (\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})]$$

(since what's inside the expectation is scalar, it equals its trace). By cyclic properties of trace, i.e. $\text{Tr}(AB) = \text{Tr}(BA)$

$$= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\text{Tr}((\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})^\top)]$$

By linearity of trace and expectation we can swap them:

$$= \text{Tr}(\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [(\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})^\top])$$

$(\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I})$ depends neither on $\mathbf{x}_0$ nor $\mathbf{Z}$ so we treat it as a constant:

$$= \text{Tr}((\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I}) \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})^\top]) \quad (1.9)$$

Note that:

$$\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0})^\top] = \Sigma_{\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}} + \mathbb{E}[\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}]\mathbb{E}[\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}]^\top$$

We have:

$$\Sigma_{\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}} = \Sigma_{p_0} + \sigma^2\mathbf{I}, \quad \mathbb{E}[\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}] = \mathbf{0}$$

So (1.9) becomes:

$$= \text{Tr}((\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I})(\Sigma_{p_0} + \sigma^2\mathbf{I})) \quad (1.10)$$

Now consider the second term in the expectation (1.7); note that since $\mathbf{Z} \perp \mathbf{x}_0$ and $\mathbb{E}\mathbf{Z} = \mathbf{0}$, the cross-terms are zeroed out:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [2((\mathbf{W}_\sigma - \mathbf{I})(\mathbf{x}_0 + \sigma\mathbf{Z} - \mu_{p_0}))^\top \sigma\mathbf{Z}] \\ &= 2\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [((\mathbf{W}_\sigma - \mathbf{I})\sigma\mathbf{Z})^\top \sigma\mathbf{Z}] \\ &= 2\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\text{Tr}(((\mathbf{W}_\sigma - \mathbf{I})\sigma\mathbf{Z})^\top \sigma\mathbf{Z})] \\ &= 2\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\text{Tr}(\sigma\mathbf{Z}(\sigma\mathbf{Z})^\top (\mathbf{W}_\sigma - \mathbf{I})^\top)] \\ &= 2\text{Tr}(\mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\sigma\mathbf{Z}(\sigma\mathbf{Z})^\top (\mathbf{W}_\sigma - \mathbf{I})^\top]) \\ &= 2\text{Tr}(\mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [\sigma\mathbf{Z}(\sigma\mathbf{Z})^\top] (\mathbf{W}_\sigma - \mathbf{I})^\top) \\ &= 2\text{Tr}(\sigma^2\mathbf{I}(\mathbf{W}_\sigma - \mathbf{I})^\top) = 2\sigma^2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) \end{aligned} \quad (1.11)$$

Now consider the third term in the expectation (1.7) is $\mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} [(\sigma\mathbf{Z})^\top (\sigma\mathbf{Z})] = \sigma^2\text{Tr}(\mathbf{I})$.

Note that $\mathbf{I} - \mathbf{W}_\sigma = \mathbf{I} - \Sigma_{p_0}(\Sigma_{p_0} + \sigma^2\mathbf{I})^{-1} = \sigma^2(\Sigma_{p_0} + \sigma^2\mathbf{I})^{-1}$. Then:

$$\begin{aligned} \mathcal{L}_\sigma &= \text{Tr}((\mathbf{W}_\sigma - \mathbf{I})^\top (\mathbf{W}_\sigma - \mathbf{I})(\Sigma_{p_0} + \sigma^2\mathbf{I})) + 2\sigma^2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) + \sigma^2\text{Tr}(\mathbf{I}) \\ &= \text{Tr}((\mathbf{W}_\sigma - \mathbf{I})^\top (-\sigma^2(\Sigma_{p_0} + \sigma^2\mathbf{I})^{-1})(\Sigma_{p_0} + \sigma^2\mathbf{I})) + 2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) + \sigma^2\text{Tr}(\mathbf{I}) \\ &= -\sigma^2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) + 2\sigma^2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) + \sigma^2\text{Tr}(\mathbf{I}) \\ &= \sigma^2\text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) + \sigma^2\text{Tr}(\mathbf{I}) \\ &= \sigma^2\text{Tr}(\mathbf{W}_\sigma) = \sigma^2\text{Tr}(\Sigma_{p_0}(\Sigma_{p_0} + \sigma^2\mathbf{I})^{-1}) \end{aligned}$$

Now let's derive a closed-form for $\mathcal{L}_{\sigma, U}$. Recall we're considering linear conditional denoiser $D(\mathbf{y}; \sigma, U) = \mathbf{W}_\sigma \mathbf{y} + \mathbf{V}_\sigma U + \mathbf{b}_\sigma$. Then:

$$\mathcal{L}_{\sigma, U} = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), U} \|\mathbf{W}_\sigma \mathbf{y} + \mathbf{V}_\sigma U + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$$

Similarly to the case of $\mathcal{L}_\sigma$, we want a closed-form expression for $\mathcal{L}_{\sigma,U}$; so we optimize for the weights $\mathbf{W}_\sigma$, $\mathbf{V}_\sigma$ and $\mathbf{b}_\sigma$. First take the derivative of $\mathcal{L}_{\sigma,U}$ w.r.t. $\mathbf{V}_\sigma$ and set to 0; note that

$$\nabla_{\mathbf{V}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 = 2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0) \mathbf{U}^\top$$

So $\|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$ continuous has its partial derivative w.r.t. $\mathbf{V}_\sigma$ exists, and $\mathbb{E}[2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0) \mathbf{U}^\top]$ is bounded if $\mathbb{E}[\mathbf{U} \mathbf{U}^\top]$ bounded, i.e. the second moment of $\mathbf{U}$ is finite, and $\mathbb{E}[\mathbf{x}_0 \mathbf{U}^\top] = \mathbb{E}[\mathbf{x}_0] \mathbb{E}[\mathbf{U}^\top] + \text{Cov}(\mathbf{U}^\top, \mathbf{x}_0)$ bounded, so first moment of $\mathbf{p}_0$ and $\mathbf{U}$ are finite and their covariance of $\mathbf{U}$ and $\mathbf{p}_0$ finite; in which case we can apply Leibniz's rule and push the differentiation inside the expectation:

$$\begin{aligned} & \nabla_{\mathbf{V}_\sigma} \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} [\nabla_{\mathbf{V}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2] \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} [2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0) \mathbf{U}^\top] \\ &= 2(\mathbb{E}_{\mathbf{x}_0 \sim p_0} [(\mathbf{W}_\sigma - \mathbf{I}) \mathbf{x}_0 \mathbf{U}^\top] + \mathbf{V}_\sigma \mathbb{E}[\mathbf{U} \mathbf{U}^\top] + \mathbf{b}_\sigma \mathbb{E}[\mathbf{U}^\top]) \\ &= 2((\mathbf{W}_\sigma - \mathbf{I})(\mathbb{E}_{\mathbf{x}_0 \sim p_0} [\mathbf{x}_0] \mathbb{E}[\mathbf{U}^\top] + \text{Cov}(\mathbf{x}_0, \mathbf{U})) + \mathbf{V}_\sigma \mathbb{E}[\mathbf{U} \mathbf{U}^\top] + \mathbf{b}_\sigma \mathbb{E}[\mathbf{U}^\top]) \\ &= 2((\mathbf{W}_\sigma - \mathbf{I})(\mu_{p_0} \mu_U^\top + \text{Cov}(\mathbf{x}_0, \mathbf{U})) + \mathbf{V}_\sigma (\Sigma_U + \mu_U \mu_U^\top) + \mathbf{b}_\sigma \mu_U^\top) \end{aligned}$$

Setting the above to 0 and solve for $\mathbf{V}_\sigma$:

$$\mathbf{V}_\sigma = ((\mathbf{I} - \mathbf{W}_\sigma)(\mu_{p_0} \mu_U^\top + \text{Cov}(\mathbf{x}_0, \mathbf{U})) - \mathbf{b}_\sigma \mu_U^\top)(\Sigma_U + \mu_U \mu_U^\top)^{-1} \quad (1.12)$$

Now take the derivative of $\mathcal{L}_{\sigma,U}$ w.r.t. $\mathbf{b}_\sigma$ and set to 0; note that

$$\nabla_{\mathbf{b}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 = 2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0)$$

So $\|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$ continuous, and its partial derivative w.r.t. $\mathbf{b}_\sigma$ exists, and $\mathbb{E}[2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0)]$ is bounded if $\mathbb{E}[\mathbf{x}_0]$ and $\mathbb{E}[\mathbf{U}]$ bounded, i.e. the first moment of $\mathbf{p}_0$ is finite; in which case again apply Leibniz's rule:

$$\begin{aligned} & \nabla_{\mathbf{b}_\sigma} \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} [\nabla_{\mathbf{b}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2] \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} [2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0)] \\ &= 2(\mathbb{E}_{\mathbf{x}_0 \sim p_0} [(\mathbf{W}_\sigma - \mathbf{I}) \mathbf{x}_0] + \mathbf{V}_\sigma \mathbb{E}[\mathbf{U}] + \mathbf{b}_\sigma) \end{aligned}$$

Setting the above to 0 and solve for $\mathbf{b}_\sigma$:

$$\mathbf{b}_\sigma = (\mathbf{I} - \mathbf{W}_\sigma) \mu_{p_0} - \mathbf{V}_\sigma \mu_U \quad (1.13)$$

Now substitute the expression for optimal $\mathbf{b}_\sigma$ in (1.13) into the expression for optimal $\mathbf{V}_\sigma$ in (1.12), in order to rewrite $\mathbf{V}_\sigma$ in terms of just $\mathbf{W}_\sigma$:

$$\begin{aligned} \mathbf{V}_\sigma &= ((\mathbf{I} - \mathbf{W}_\sigma)(\mu_{p_0} \mu_U^\top + \text{Cov}(\mathbf{x}_0, \mathbf{U})) - ((\mathbf{I} - \mathbf{W}_\sigma) \mu_{p_0} - \mathbf{V}_\sigma \mu_U) \mu_U^\top)(\Sigma_U + \mu_U \mu_U^\top)^{-1} \\ &= (\mathbf{I} - \mathbf{W}_\sigma)(\text{Cov}(\mathbf{x}_0, \mathbf{U}) + \mathbf{V}_\sigma \mu_U \mu_U^\top)(\Sigma_U + \mu_U \mu_U^\top)^{-1} \end{aligned}$$

So:

$$\begin{aligned} \mathbf{V}_\sigma (\mathbf{I} - \mu_U \mu_U^\top)(\Sigma_U + \mu_U \mu_U^\top)^{-1} &= (\mathbf{I} - \mathbf{W}_\sigma) \text{Cov}(\mathbf{x}_0, \mathbf{U})(\Sigma_U + \mu_U \mu_U^\top)^{-1} \\ \mathbf{V}_\sigma \Sigma_U (\Sigma_U + \mu_U \mu_U^\top)^{-1} &= (\mathbf{I} - \mathbf{W}_\sigma) \text{Cov}(\mathbf{x}_0, \mathbf{U})(\Sigma_U + \mu_U \mu_U^\top)^{-1} \\ \mathbf{V}_\sigma &= (\mathbf{I} - \mathbf{W}_\sigma) \text{Cov}(\mathbf{x}_0, \mathbf{U}) \Sigma_U^{-1} \end{aligned} \quad (1.14)$$

Recall that the regression coefficient of $\mathbf{x}_0$ onto $\mathbf{y}$ is $\frac{\text{Cov}(\mathbf{x}, \mathbf{y})}{\text{Var}(\mathbf{y})}$, so $\mathbf{V}_\sigma$ is the regression coefficient of $(\mathbf{I} - \mathbf{W}_\sigma) \mathbf{x}_0$ onto $\mathbf{U}$, which makes sense since $\mathbf{U}$ is being used to predict the rest of $\mathbf{x}_0$ that $\mathbf{W}_\sigma$ has not yet explained. With that, the closed-form for $\mathbf{b}_\sigma$ in terms of $\mathbf{W}_\sigma$ is:

$$\begin{aligned} \mathbf{b}_\sigma &= (\mathbf{I} - \mathbf{W}_\sigma) \mu_{p_0} - (\mathbf{I} - \mathbf{W}_\sigma) \text{Cov}(\mathbf{x}_0, \mathbf{U}) \Sigma_U^{-1} \mu_U \\ &= (\mathbf{I} - \mathbf{W}_\sigma)(\mu_{p_0} - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \Sigma_U^{-1} \mu_U) \end{aligned} \quad (1.15)$$

Next we find the optimal $\mathbf{W}_\sigma$. Take derivative of $\mathcal{L}_{\sigma,U}$ w.r.t. $\mathbf{W}_\sigma$ and set to 0; similarly we note:

$$\nabla_{\mathbf{W}_\sigma} \|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 = 2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0)(\mathbf{x}_0 + \sigma \mathbf{Z})^\top$$

So $\|\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$ continuous, and its partial derivative w.r.t. $\mathbf{W}_\sigma$ exists, and $\mathbb{E}[2(\mathbf{W}_\sigma(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{V}_\sigma \mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0)(\mathbf{x}_0 + \sigma \mathbf{Z})^\top]$ is bounded if $\mathbb{E}[\mathbf{x}_0 \mathbf{x}_0^\top]$ is bounded, i.e. the second moment of $\mathbf{p}_0$ is finite, and $\mathbb{E}[\mathbf{U} \mathbf{x}_0^\top] = \mathbb{E}[\mathbf{U}] \mathbb{E}[\mathbf{x}_0^\top] +$

$\text{Cov}(U, x_0^\top)$ bounded, i.e. the first moment of U and the covariance of U and p_0 also finite; in which case again apply Leibniz's rule:

$$\begin{aligned} & \nabla_{W_\sigma} \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} \|W_\sigma(x_0 + \sigma Z) + V_\sigma U + b_\sigma - x_0\|_2^2 \\ &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I)} [\nabla_{W_\sigma} \|W_\sigma(x_0 + \sigma Z) + V_\sigma U + b_\sigma - x_0\|_2^2] \end{aligned} \quad (1.16)$$

$$\begin{aligned} &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I)} [2(W_\sigma(x_0 + \sigma Z) + V_\sigma U + b_\sigma - x_0)(x_0 + \sigma Z)^\top] \\ &= 2(W_\sigma \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I)} [(x_0 + \sigma Z)(x_0 + \sigma Z)^\top] \\ &\quad + b_\sigma \mathbb{E}[x_0^\top] - \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I)} [x_0(x_0 + \sigma Z)^\top] + \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I)} [V_\sigma U(x_0 + \sigma Z)^\top]) \\ &= 2(W_\sigma(\mathbb{E}_{x_0 \sim p_0} [x_0 x_0^\top] + \sigma^2 I) - \mathbb{E}_{x_0 \sim p_0} [x_0 x_0^\top] + b_\sigma \mathbb{E}[x_0]^\top + V_\sigma \mathbb{E}[U x_0^\top]) \end{aligned} \quad (1.17)$$

$$\begin{aligned} & \text{(since } \mathbb{E}_{Z \sim \mathcal{N}(0, I)} [ZZ^\top] = I \text{ and } Z \perp x_0, Z \perp U) \\ &= 2(W_\sigma(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I) - (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top) + b_\sigma \mu_{p_0}^\top + V_\sigma(\mu_U \mu_{p_0}^\top + \text{Cov}(x_0, U))) \end{aligned} \quad (1.18)$$

Setting the above to 0 and solve for W_σ :

$$W_\sigma = ((\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top) - b_\sigma \mu_{p_0}^\top - V_\sigma(\mu_U \mu_{p_0}^\top + \text{Cov}(x_0, U)))(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1}$$

Substitute V_σ from (1.14) and b_σ from (1.15) to get:

$$\begin{aligned} W_\sigma &= (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top)(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} - (b_\sigma \mu_{p_0}^\top + V_\sigma(\mu_U \mu_{p_0}^\top + \text{Cov}(x_0, U)))(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ W_\sigma &= I - \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ &\quad - ((I - W_\sigma)(\mu_{p_0} - \text{Cov}(x_0, U) \Sigma_U^{-1} \mu_U) \mu_{p_0}^\top + (I - W_\sigma) \text{Cov}(x_0, U) \Sigma_U^{-1} (\mu_U \mu_{p_0}^\top + \text{Cov}(x_0, U))) \\ &\quad (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ &\quad + (I - W_\sigma)((\mu_{p_0} - \text{Cov}(x_0, U) \Sigma_U^{-1} \mu_U) \mu_{p_0}^\top + \text{Cov}(x_0, U) \Sigma_U^{-1} (\mu_U \mu_{p_0}^\top + \text{Cov}(x_0, U))) (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} + (I - W_\sigma)(\mu_{p_0} \mu_{p_0}^\top + \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U)) (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} (I - (\mu_{p_0} \mu_{p_0}^\top + \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U)) (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1})^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} ((\Sigma_{p_0} + \sigma^2 I - \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U)) (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1})^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I)^{-1} (\Sigma_{p_0} + \mu_{p_0} \mu_{p_0}^\top + \sigma^2 I) (\Sigma_{p_0} + \sigma^2 I - \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U))^{-1} \\ I - W_\sigma &= \sigma^2(\Sigma_{p_0} + \sigma^2 I - \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U))^{-1} \\ W_\sigma &= I - \sigma^2(\Sigma_{p_0} + \sigma^2 I - \text{Cov}(x_0, U) \Sigma_U^{-1} \text{Cov}(x_0, U))^{-1} \end{aligned}$$

Now let's derive the closed-form of $\mathcal{L}_{\sigma, U}$:

$$\begin{aligned} \mathcal{L}_{\sigma, U} &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} \|W_\sigma(x_0 + \sigma Z) + V_\sigma U + b_\sigma - x_0\|_2^2 \\ \mathcal{L}_{\sigma, U} &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} \|(W_\sigma - I)(x_0 + \sigma Z) + V_\sigma U + b_\sigma + \sigma Z\|_2^2 \end{aligned}$$

Letting the shorthand $B = \text{Cov}(x_0, U) \Sigma_U^{-1} \mu_U$. Substituting:

$$\begin{aligned} \mathcal{L}_{\sigma, U} &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} \|(W_\sigma - I)(x_0 + \sigma Z) + (I - W_\sigma) \text{Cov}(x_0, U) \Sigma_U^{-1} U + (I - W_\sigma)(\mu_{p_0} - B) + \sigma Z\|_2^2 \\ &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} \|(W_\sigma - I)(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B) + \sigma Z\|_2^2 \\ &= \mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} [(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)^\top (W_\sigma - I)^\top \\ &\quad (W_\sigma - I)(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B) \\ &\quad + 2(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)^\top (W_\sigma - I)^\top (\sigma Z) + (\sigma Z)^\top (\sigma Z)] \end{aligned}$$

Consider the first expectation terms; notice that $(W_\sigma - I)$ should already be known, so we only have to consider the expectation of the inner-product:

$$\mathbb{E}_{x_0 \sim p_0, Z \sim \mathcal{N}(0, I), U} [(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)^\top (W_\sigma - I)^\top (W_\sigma - I)(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)] \quad (1.19)$$

$$= \text{Tr}((W_\sigma - I)^\top (W_\sigma - I) \mathbb{E}_{x_0, Z, U} [(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)(x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B)^\top]) \quad (1.20)$$

We have:

$$\begin{aligned} & x_0 + \sigma Z - \text{Cov}(x_0, U) \Sigma_U^{-1} U - \mu_{p_0} + B \\ &= (x_0 + \sigma Z - \mu_{p_0}) - \text{Cov}(x_0, U) \Sigma_U^{-1} (U - \mu_U) \end{aligned}$$

So the expectation in (1.20) becomes:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [((\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}) - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U)) ((\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}) - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U))^\top] \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})^\top - 2(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}) \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U)^\top \\ &+ \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U))^\top] \end{aligned} \quad (1.21)$$

We have:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})^\top] \\ &= (\boldsymbol{\Sigma}_{\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}}) + \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})] \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})]^\top \\ &= (\boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I}) \end{aligned}$$

This is because $\mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0})] = \mathbf{0}$ and $\boldsymbol{\Sigma}_{\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}} = \boldsymbol{\Sigma}_{\mathbf{x}_0 - \boldsymbol{\mu}_{p_0}} + \boldsymbol{\Sigma}_{\sigma \mathbf{Z}} = \boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I}$ (since $\mathbf{Z} \perp \mathbf{x}_0$). And:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{U}} [(\mathbf{x}_0 + \sigma \mathbf{Z} - \boldsymbol{\mu}_{p_0}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U))^\top] \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{U}} [(\mathbf{x}_0 - \boldsymbol{\mu}_{p_0}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U))^\top] \quad (\text{since } \mathbf{Z} \perp \mathbf{U} \text{ and } \mathbb{E} \mathbf{Z} = \mathbf{0}) \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{U}} [(\mathbf{x}_0 - \boldsymbol{\mu}_{p_0}) (\mathbf{U} - \boldsymbol{\mu}_U)^\top] (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \\ &= \text{Cov}(\mathbf{x}_0, \mathbf{U}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \end{aligned}$$

Also:

$$\begin{aligned} & \mathbb{E}_{\mathbf{U}} [\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} (\mathbf{U} - \boldsymbol{\mu}_U))^\top] \\ &= \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} \mathbb{E}_{\mathbf{U}} [(\mathbf{U} - \boldsymbol{\mu}_U) (\mathbf{U} - \boldsymbol{\mu}_U)^\top] (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \\ &= \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} \boldsymbol{\Sigma}_U (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \\ &= \text{Cov}(\mathbf{x}_0, \mathbf{U}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \end{aligned}$$

(since $\boldsymbol{\Sigma}_U^{-1}$ is symmetric and $\text{Cov}(\mathbf{x}_0, \mathbf{U})^\top = \text{Cov}(\mathbf{U}, \mathbf{x}_0)$). So (1.21) becomes:

$$\begin{aligned} & \boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I} - 2 \text{Cov}(\mathbf{x}_0, \mathbf{U}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top + \text{Cov}(\mathbf{x}_0, \mathbf{U}) (\text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1})^\top \\ &= \boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I} - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} \text{Cov}(\mathbf{x}_0, \mathbf{U}) \end{aligned}$$

Let $\mathbf{S} = \boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I} - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} \text{Cov}(\mathbf{x}_0, \mathbf{U})$. Then $\mathbf{I} - \mathbf{W} = \sigma^2 \mathbf{S}^{-1}$. And then:

$$\text{Tr}((-\sigma^2 \mathbf{S}^{-1})^\top (-\sigma^2 \mathbf{S}^{-1}) \mathbf{S}) = \sigma^4 \text{Tr}(\mathbf{S}^{-1})$$

Now consider the second expectation; since $\mathbf{Z} \perp \mathbf{U}$ and $\mathbf{Z} \perp \mathbf{x}_0$ so we can decompose the expectation of product into product of expectation, and $\mathbb{E} \mathbf{Z} = \mathbf{0}$ so the cross-terms are all zeroed out, leaving:

$$\begin{aligned} & \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [(\sigma \mathbf{Z})^\top (\mathbf{W}_\sigma - \mathbf{I})^\top (\sigma \mathbf{Z})] \\ &= \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [((\mathbf{W}_\sigma - \mathbf{I})(\sigma \mathbf{Z}))^\top (\sigma \mathbf{Z})] = \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [(\sigma \mathbf{Z})^\top (\mathbf{W}_\sigma - \mathbf{I})(\sigma \mathbf{Z})] \\ &= \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\text{Tr}((\sigma \mathbf{Z})^\top (\mathbf{W}_\sigma - \mathbf{I})(\sigma \mathbf{Z}))] \\ &= \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\text{Tr}((\mathbf{W}_\sigma - \mathbf{I})(\sigma \mathbf{Z})(\sigma \mathbf{Z})^\top)] \\ &= \text{Tr}((\mathbf{W}_\sigma - \mathbf{I}) \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [(\sigma \mathbf{Z})(\sigma \mathbf{Z})^\top]) \\ &= \sigma^2 \text{Tr}(\mathbf{W}_\sigma - \mathbf{I}) = -\sigma^4 \text{Tr}(\mathbf{S}^{-1}) \end{aligned}$$

Finally the third expectation terms is just $\mathbb{E}[(\sigma \mathbf{Z})^\top (\sigma \mathbf{Z})] = \sigma^2 \text{Tr}(\mathbf{I})$. Hence:

$$\begin{aligned} \mathcal{L}_{\sigma, U} &= \sigma^4 \text{Tr}(\mathbf{S}^{-1}) + 2(-\sigma^4 \text{Tr}(\mathbf{S}^{-1})) + \sigma^2 \text{Tr}(\mathbf{I}) \\ &= \sigma^2 \text{Tr}(\mathbf{I}) - \sigma^4 \text{Tr}(\mathbf{S}^{-1}) \\ &= \sigma^2 \text{Tr}(\mathbf{I} - \sigma^2 \mathbf{S}^{-1}) \end{aligned}$$

Note that $\mathbf{I} - \sigma^2 \mathbf{S}^{-1} = \mathbf{W}_\sigma$, so really $\mathcal{L}_{\sigma, U} = \sigma^2 \text{Tr}(\mathbf{W}_\sigma)$; recall that in the unconditional case, $\mathcal{L}_\sigma = \sigma^2 \text{Tr}(\mathbf{W}_\sigma^{\text{uncond}})$ also, so our derivations are consistent. Furthermore:

$$\begin{aligned} \mathcal{L}_\sigma - \mathcal{L}_{\sigma, U} &= \sigma^2 \text{Tr}(\mathbf{I} - \sigma^2 (\boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I})^{-1} - (\mathbf{I} - \sigma^2 \mathbf{S}^{-1})) \\ &= \sigma^4 \text{Tr}((\boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I} - \text{Cov}(\mathbf{x}_0, \mathbf{U}) \boldsymbol{\Sigma}_U^{-1} \text{Cov}(\mathbf{x}_0, \mathbf{U}))^{-1} - (\boldsymbol{\Sigma}_{p_0} + \sigma^2 \mathbf{I})^{-1}) \end{aligned}$$

So using parametrization $t = \sigma^2$:

$$\begin{aligned}
I(X, U) &\approx \frac{1}{2} \int_0^\infty \frac{dt}{t^2} (\text{MMSE}[X | Y_t] - \text{MMSE}[X | Y_t, U]) \\
&\approx \frac{1}{2} \int_0^\infty \frac{d\sigma}{\sigma^3} (\text{MMSE}[X | Y_\sigma] - \text{MMSE}[X | Y_\sigma, U]) \\
&\approx \frac{1}{2} \int_0^\infty \sigma \text{Tr}((\Sigma_{p_0} + \sigma^2 \mathbf{I} - \text{Cov}(\mathbf{x}_0, U) \Sigma_U^{-1} \text{Cov}(\mathbf{x}_0, U))^{-1} - (\Sigma_{p_0} + \sigma^2 \mathbf{I})^{-1}) d\sigma
\end{aligned}$$

(Note that in the Gaussian case the approximation is really equality). Let $\mathbf{C} = \text{Cov}(\mathbf{x}_0, U) \Sigma_U^{-1} \text{Cov}(\mathbf{x}_0, U)$. Consider:

$$\begin{aligned}
I(X, U) &= \frac{1}{2} \int_0^\infty \frac{dt}{t^2} (\text{MMSE}[X | Y_t] - \text{MMSE}[X | Y_t, U]) \\
&= \frac{1}{2} \int_0^\infty \frac{dt}{t^2} t \text{Tr}((\Sigma_{p_0} + t\mathbf{I} - \mathbf{C})^{-1} - (\Sigma_{p_0} + t\mathbf{I})^{-1})
\end{aligned}$$

Let Σ_{p_0} and $\mathbf{C}$ be simultaneously diagonalizable i.e. sharing the same change of basis matrix, which is the case for jointly Gaussian, with eigenvalue λ_i and c_i respectively. In fact, for jointly Gaussian $\mathbf{x}_0$ and U , $\text{Cov}(\mathbf{x}_0 | U) = \Sigma_{p_0} - \text{Cov}(\mathbf{x}_0, U) \Sigma_U^{-1} \text{Cov}(\mathbf{x}_0, U)$.

Then $\text{Tr}(\Sigma_{p_0} + t\mathbf{I} - \mathbf{C})^{-1} = \sum \frac{1}{\lambda_i + t - c_i}$ and $(\Sigma_{p_0} + t\mathbf{I})^{-1} = \sum \frac{1}{\lambda_i + t}$.

$$I(X, U) = \frac{1}{2} \int_0^\infty \frac{dt}{t} \left(\sum \frac{1}{\lambda_i + t - c_i} - \frac{1}{\lambda_i + t} \right)$$